Off Earth Landing and Launch Pad Construction—A Critical Technology for Establishing a Long-Term Presence on Extraterrestrial Surfaces

Nathan J. Gelino[1]; Robert P. Mueller[2]; Robert W. Moses, Ph.D.[3]; James G. Mantovani, Ph.D.[4]; Philip T. Metzger, Ph.D. [5]; Brad C. Buckles[6]; and Laurent Sibille, Ph.D.[7]

[1]Swamp Works, Exploration and Research Technologies, Kennedy Space Center, National Aeronautics and Research Administration, KSC, FL
[2]Swamp Works, Exploration and Research Technologies, Kennedy Space Center, National Aeronautics and Research Administration, KSC, FL
[3]Langley Research Center, National Aeronautics and Space Administration, Hampton, VA
[4]Swamp Works, Exploration and Research Technologies, Kennedy Space Center, National Aeronautics and Research Administration, KSC, FL
[5]Florida Space Institute, Univ. of Central Florida, Orlando, FL
[6]Swamp Works, The Bionetics Corp., Kennedy Space Center, FL
[7]Swamp Works, Southeastern Universities Research Association, Kennedy Space Center, FL

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ABSTRACT

In 2017, Space Policy Directive–1 was issued to ensure that "the United States will lead the return of humans to the Moon for long-term exploration and utilization" and engage the National Aeronautics and Space Administration (NASA) in the national effort. A long-term presence on the Moon will require numerous lunar landings and launches to build up surface assets, rotate crew, and deliver logistics supplies to/from the Moon. Interactions between landing/launch rocket engine plumes and the unprepared lunar surface will lead to erosion and ejection of regolith particles at high velocities from beneath the vehicle. The associated ejecta elevates mission risks by: obscuring sensors and human vision during landing via lofted dust; cratering/modifying the surface that the vehicle will land upon; and subjecting the vehicle, surrounding assets, and potentially orbital assets to impacts and contamination from high-velocity dust particles. The construction of a reusable landing/launch pad infrastructure is a method to mitigate such risks. A landing/launch pad system can provide a known, benign and reliable landing/launch surface, minimize regolith ejecta, and protect surrounding assets from liberated particles. Associated blast barriers can provide protection in case of a landing/launch catastrophic anomaly. This paper summarizes previous work towards construction of off-Earth landing/launch pad infrastructure and begins to identify key technology gaps. Additionally, this paper establishes metrics for comparison of pad construction technologies to guide future trade studies. The overall objective of this paper is to baseline the state of the art of off-Earth landing/launch pad construction technologies and serve as the starting point for further technical development.

INTRODUCTION

Launch and landing of spacecraft on Earth is a well-defined discipline with over seventy years of operational experience to inform and optimize the subsystems that perform the required functions. When all of these functions are combined into one overall system, the resulting infrastructure is called a Launch & Landing Complex (LLC). Recently, terrestrial vertical

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landing operations adopted for commercial rocket first stages returning to the launch site [‘VTVL’, (2020)] has expanded the infrastructure required in a LLC, to include landing pads which mitigate risk by providing a safe and reliable landing zone. A terrestrial LLC can be a complex system with an extensive footprint and functions including structural supports, rocket plume exhaust management, Foreign Object Debris (FOD) ejecta prevention and mitigation, fluid, power and communication services. On-site system verification and site safety services aim to protect surface assets and personnel from detonation, conflagration, fragmentation, overpressure or toxic hazards. Current NASA plans include landing and launching large, human rated spacecraft on multiple missions to the Moon and Mars. Beyond the concerns of the first landing, a functionally equivalent LLC will have to be developed for repeated returns to one or a few sites for a sustained presence, and ensure reliable, safe operations at reasonable life cycle costs. However, this paper will focus only on the technologies and methods needed to construct the Launch & Landing pad (LLP) and associated blast barriers of an Off –Earth LLC, for a vertical landing and takeoff vehicle.

Figure 1: Launch Complex 39A and 39B at the NASA Kennedy Space Center during the Space Shuttle Program (Source: NASA)

Figure 2: SpaceX inc. Falcon 9 Rocket First Stage Landing on a Terrestrial Landing Pad at Cape Canaveral (Source: Wikipedia)

When a lander vehicle launches or lands on an extraterrestrial body, the rocket engine exhaust plume impinges on the surface and interacts with the regolith to create blast ejecta and associated cratering of the surface. Lunar regolith blast ejecta travels at high velocities (>2,000 m/s) for long distances (kilometers) in a vacuum environment (Lane, 2015) creating hazards for surrounding assets and it can also impact the bottom of the lander vehicle, risking damage to the engines, thermal insulation and sensors. Ballistic particles can possibly enter cislunar space and achieve orbit as debris, if the ejecta is sufficiently energetic. The cratering and regolith erosion can endanger the vehicle itself by affecting the soil stability under the landing gear. Landing on unpredictable terrain with varying topography, natural craters and rock hazards is also hazardous risks tipping a lander at dangerous angles in extreme conditions that may also violate maximum slope angles for subsequent launch operations. During launch, an overpressure pulse created by the ignition of the rocket engines can pose significant ejecta risks to the vehicle. Dust clouds raised during landing limit the efficacy of sensors and reduce visibility for the astronaut pilots, creating significant real-time risk during landing site selection by the pilot or computer navigation system. Future lunar and Martian spaceports will require mitigations to these launch and landing risks (Benaroya, 2002).

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Figure 3. NASA Morpheus lander at Kennedy Space Center showing rocket plume impingement

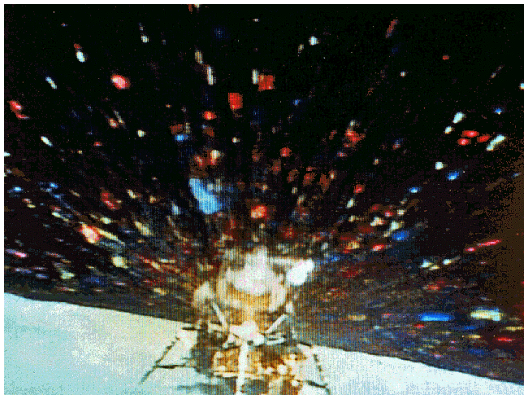

Figure 4. Apollo 16 ascent stage liftoff showing extensive ejecta debris (NASA PHOTO: s72-35613.)

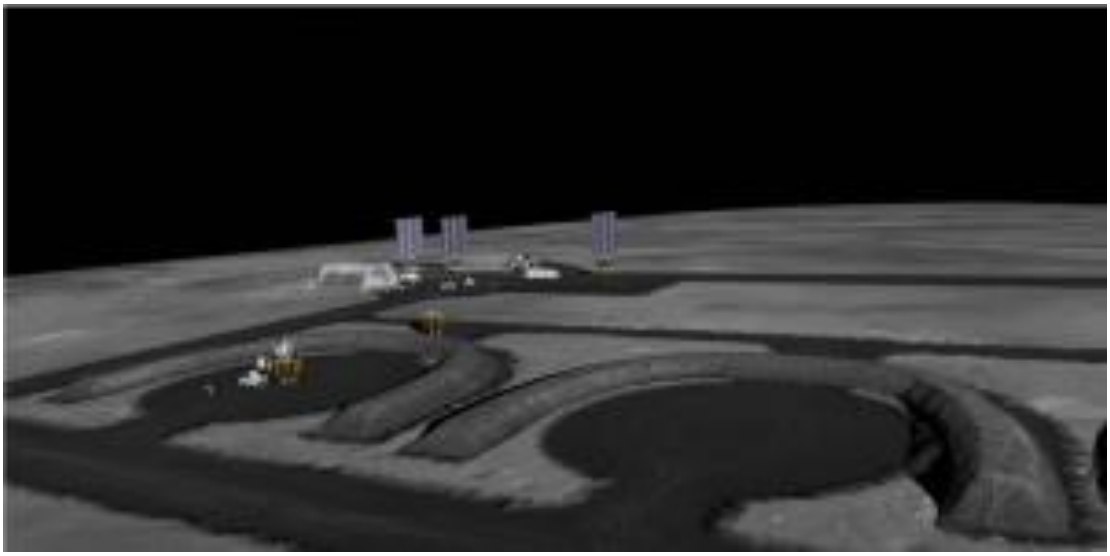

Figure 5. Concept for a Lunar Launch and Landing Complex (LLC)
Source: NASA (artist: Maxwell)

MOTIVATION

There are four main objectives of this paper: 1) To establish the state of the art in LLP construction methodologies. 2) To propose criteria for trade studies of LLP concepts. 3) To publicly share the authors' ideas for potential LLP solutions. 4) To serve as the starting point for future development of LLP technologies.

Establishing the state of the art in the area of off-Earth Launch & Landing Pad concepts will baseline the work that has been completed thus far and highlight the wide span between current Technology Readiness Levels (TRLs) and operational readiness. The authors aim to communicate the need for funding in this area in the near term by illustrating that there is much work to be completed before a truly viable option exists.

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Setting forth criteria for trade studies of LLP concepts is important for several reasons. The first and most straightforward is to establish a framework for performing trade studies on LLP concepts. This will enable NASA to select the most promising concepts for continued development, and it will also help technology developers understand how their concepts compare with others and the priorities of development efforts.

Publicly sharing Swamp Work's ideas for LLP solutions serves two purposes. The first is providing insight into the concepts that are being considered internally at NASA. If concepts exist that are not under consideration, we would like to be informed of them. Secondly, presenting LLP concepts in this way will broaden researchers' perspectives on the topic, encourage public debate, trigger new ideas, and stimulate innovation.

The trade study framework presented in this paper is intended to be the starting point for future development. It is anticipated that quantifiable data points will be established for each of the most promising construction concepts. This serves to inform trades, and it also clearly identifies where gaps in the research exist. Data gaps will serve as way markers for prioritizing future funding opportunities.

PRELIMINARY ENGINEERING ASSESSMENT

In 2019, the NASA Artemis lunar exploration program outlined a phasing plan of lunar lander missions that targets 2024 as the year when humans will return to the surface of the Moon under the Human Landing System (HLS) program. HLS landers will target the lunar South Pole region initially and will contribute to the establishment of a surface infrastructure for a sustained phase beginning in 2028.

Prior to 2024, NASA's Commercial Lunar Payload Services (CLPS) program plans early missions to deliver science and technology payloads of up to 300 kg total mass to the lunar surface using small and mid-class landers. The CLPS landers typically have multiple descent engines that can provide a combined thrust of <4000 N during powered descent.

Beyond the smaller scale CLPS landers, the large-scale non-human HLS landers are expected to deliver at least 12 t of payload to the lunar surface and require descent engines with 100 kN of thrust during powered descent. By contrast, the NASA Apollo lunar lander module descent engine had a thrust of 16 kN. Lane and Metzger, 2015 estimated that the Apollo descent engine displaced approximately 2.6 metric tons (t) of regolith during landing. Terrestrial experiments showed that, for the case of non-rarefied plumes, erosion rate is proportional to thrust, which is proportional to mass of the lander (Metzger, 2010). Vacuum chamber experiments showed that rarefied flow scales faster with thrust than non-rarefied cases (Metzger, 2016), and analysis of Apollo Lunar Module videos showed that erosion rate scaled as thrust (or lander mass) to the 2.5 power [Lane and Metzger, 2015]. However, it is not known whether the density of plume under a much larger lander is enough to transition from the 2.5 power to the direct proportionality of non-rarefied cases. If we assume that the mass erosion rate is approximately proportional to the lander mass, then a HLS lander may displace a total mass of 16 t of lunar regolith during landing on unmodified surfaces. This ejecta will travel at velocities exceeding lunar escape velocity (Lane, 2012) and can cause severe hardware damage to any assets located in its path (Immer, 2011).

Novel lunar lander configurations can potentially reduce the detrimental effects of plume surface interactions very significantly. The Hercules lander design proposed by NASA Langley Research Center enables powered descent to the lunar surface using multiple thrusters mounted

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at the top of the vehicle instead of only relying on a main descent engine approximately 1 meter off the ground (Komar, 2017).

NASA Marshall Space Flight Center (MSFC) is currently leading a NASA Game Changing Development project with plans to conduct ground tests during 2020-2024 of hot fire plume gas on lunar regolith soil beds under vacuum conditions which will yield valuable test data that can be used to improve computational modeling of Plume Surface Interactions. These computational models will then be used to estimate the erosion rate for any future lunar lander design with higher confidence than ever before.

STATE OF THE ART AND PAST WORK

Although many concepts of LLP construction have been investigated, none have reached a high Technology Readiness Level. The overall TRL level of the technology is approximately TRL-2. Much of the research has focused on materials development and strength characterization. Examples of research work focused on construction methods, emplacement systems, and performance of structures under simulated launch/landing conditions are summarized below.

A Vertical Take Off Vertical Landing (VTVL) pad was robotically constructed from sintered basalt pavers by the Swamp Works and Pacific International Space Center for Exploration Systems (PISCES) teams in Hawaii (Kelso, 2016). The project demonstrated construction and testing of a pad using only materials available in situ. The ability of the pad to withstand the rocket exhaust of a lander was tested by firing a vertical Class M solid propellant engine above the pad. The subsequent lifting of a few individual pavers by the gas was possibly due to the weakness of the seams between pavers or by cracking of the pavers under high temperature and pressure. Further work was performed to solve this by fusing pavers together with a thermite material, but further firing tests was not performed (Ferguson, 2018).

Van Susante and Metzger (2016) investigated various soil stabilization techniques including pads made from rocks and pavers on which they simulated the effects of rocket plumes with rocket nozzles and compressed air. Loose regolith and compacted regolith were not effective at mitigating erosion, while rock beds were somewhat more effective.

A lunar concrete landing pad made with KOHLS-1/KAUMLS Korean Lunar Simulants, polymer binders, and glass fiber was constructed at atmospheric pressure and tested for compression strength. Rocket plume effects were not tested on the pad (Lee, 2012).

TRADE STUDY CRITERIA

One of the primary objectives of this paper is to establish key criteria for use in comparing Launch/Landing Pad (LLP) concepts and assessing suitability for specific surface architectures. Additionally, these criteria are intended to provide guidance to future LLP technology development efforts by identifying what concept attributes should be explored and quantified when possible. The vision is to eventually produce a trade matrix with data supporting the performance of each concept in each of the criterion. Trade matrix data gaps in high potential LLP concepts will be considered higher priority for future funding. It should be noted that specific architectures will weigh each criterion differently when comparing concepts. Lander size/configuration, number of required launches/landings, location and other factors could dramatically change the requirements for LLPs.

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The construction process is divided into 3 phases: 1) Preparation and Staging, 2) Construction and 3) Operations and Maintenance. Table 1 displays each phase and its list of criteria. The Preparation and Staging Phase includes all preliminary work necessary to begin construction. Examples include mining/excavation, comminution, beneficiation, processing raw materials into construction feedstock, surface preparation, staging of materials and systems, and others. One of the most impactful trade is the origin of the materials (Earth based vs. Moon based), the associated up-mass, and the condition of the materials upon arrival at the construction site. The ratio of imported/lunar components will define the amount of power and the equipment necessary for placement into service. This includes sourcing the feedstock from lunar resources as well as the power to convert the feedstock into the end product.

Table 1. Key Trade Study Criteria for LLP Concepts

Preparation and Staging Phase	Construction Phase	Operations and Maintenance Phase
Up-Mass of Construction Materials and Systems	Constructability	Performance as a Landing/Launch Surface
Difficulty of Insitu Materials Collection, Handling, and Processing	Versatility	Expected Life
Effort of Site Preparation/Staging Time	Construction Time	Ease of Repairability
Reliance on other Surface Assets	Reliance on other Surface Assets	Reliance on Lunar, Gateway and Earth Crew Interaction
Reliance on Lunar, Gateway and Earth Crew Interaction	Reliance on Lunar, Gateway and Earth Crew Interaction	Robotics and Autonomy
Robotics and Autonomy	Robotics and Autonomy	Required Power
Robustness of process	Robustness of process	Lifecycle cost
Current Technology Readiness Level	Current Technology Readiness Level	
Required Power	Required Power	
	Ability to Verify As-Built Performance	

The Construction Phase describes the performance of work with construction materials with the objective of emplacing them in a structure ready for operations. This may require applying thermal or mechanical energy to transform or reshape the materials, laying them in geometric patterns according to LLP design, containment/stabilization, or a host of other techniques.

The Operations and Maintenance Phase begins once the LLP is placed into service. This phase continues until the site is decommissioned. Operations include launching/landing, ground transport of lander stages and whole landers, staging and use of ground support equipment, loading/unloading of crew and cargo and others. Maintenance is aimed at keeping the structures in proper working condition, which may involve resurfacing and repairs.

An additional discussion of each criterion is provided below. Some criteria are included in several phases of the construction process. In a trade matrix, each criterion below represents a column.

Up-Mass of Construction Materials and Systems: A major trade in any space endeavor is how much mass can be sent to the desired destination. Although reusable rocket stages are driving down launch costs (Richardson, 2018), the magnitude of launch/landing pad mass requirements will likely drive the use of in-situ materials. Similarly, the launch of traditional terrestrial systems capable of handling and emplacing structures on the scale of a LLP is not practical due to their mass. Concepts must minimize the launch mass requirements of construction materials and systems to be considered a viable option unless payload capacity to the lunar surface or launch rate dramatically increases.

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Difficulty of In Situ Materials Collection, Handling, and Processing: If in-situ materials are used, there is a wide spectrum of difficulty of materials collection, handling and processing. For example, a LLP concept may be able to manage with very little processing such as landing/launching on the surface as is or compacting regolith to stabilize it. Other concepts may rely on some quantity of excavated regolith in a specific size fraction, which implies excavators and comminution equipment. Still others may be significantly more challenging by requiring specific minerals or even products like ferroalloys from processes such as molten regolith electrolysis. Concepts with the least challenging materials collection, handling and processing requirements will perform the best in this criterion. Potential quantifiable metrics include mass of required resources, processing rates and yield, distance to required resources, time, and power needs.

Effort of Site Preparation/Staging Time: Various approaches to site preparation and staging exist that span a wide range of effort. Construction sites may require removal of overburden, grading, compaction, obstacle removal, provision of access, or numerous other operations. In addition, materials originating from either Earth or the Moon may need work completed on them and need to be prepared prior to their transformation into a usable construction element. For example, it may be necessary to produce and stockpile specific regolith prior to the production of sintered regolith pavers before emplacing the final structure. Potential quantifiable metrics include mass of processed resources, preparation rates, distance to required resources, time and power needs.

Reliance on other Surface Assets: Some construction concepts will require interfaces with other existing surface assets in order to complete tasks during each phase. Examples include power sourcing from fission surface power (FSP) units, transportation and mobility services from the All-Terrain Hex-Limbed Extra-Terrestrial Explorer (ATHLETE) robot, excavation by the Regolith Advanced Surface Systems Operations Robot (RASSOR) robot, etc. Reliance on other systems are not necessarily a negative factor if the assets are available. A potential quantifiable metric is the time demand on the other asset.

Reliance on Lunar, Gateway and Earth Crew Interaction: Concepts will have varying needs for humans in the loop during each phase. As with other surface assets, reliance on humans to engage in construction tasks may not be a negative factor so long as personnel are available to support. The time frame for crew support is a quantifiable metric in this area.

Robotics and Autonomy: This criterion requires an examination of whether the tasks to be performed are suitable for autonomous operations when robotics and autonomy are either required or can be considered. Additionally, particular attention must be paid to the level of maturity of existing hardware and control algorithms.

Robustness of Process: In this context, robustness refers to the ability of system/process to handle variability and remain effective to demonstrate resiliency. Robustness is challenging to quantify, but an important factor for this trade. Since so little is known about lunar surface operations with regolith, there will be conditions that were not anticipated or modeled in the design of the system. The lunar surface is an extremely challenging environment where dynamic conditions exist. Such examples include the changing illumination angles over days in constant vacuum that create shifting thermal gradients on materials and machinery while meteoritic bombardment is unpredictable by nature. The ability of a concept to adapt to known changes and unknowns and remain successful in each of the three phases through its inherent flexibility is critical.

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Current Technology Readiness Level: The TRL of a system can be quantified using the NASA TRL scale available to the public.

Required Power: Several power systems are presently considered for lunar surface operations in the context of the Artemis Program. Power will likely be required at each of the three phases of operations of a LLP. A key quantifiable trade metric is how much power is required, how the power will be produced, and the form of the delivered power (thermal, electrical, pneumatic).

Constructability: Constructability is a novel term that refers to the complexity of emplacement systems, processing, and operations. In loose terms, it means how easily the LLP can be built.

Versatility: Although the focus is on the construction of LLPs, the versatility of the construction system and processes is important. It is advantageous if the concept can be applied to other infrastructure construction projects such as dust mitigation, roads for trafficability, radiation protection, micrometeoroid protection, thermal insulation, pressurized/unpressurized structures, access through challenging terrain, and others. Items of interest are max build envelope, mobility vs. static emplacement strategies, system lifespan, horizontal vs. vertical construction capability, and others.

Construction Time: Construction time refers to how long it takes to complete construction tasks.

Ability to Verify As-Built Performance: On Earth, building code compliance is generally confirmed by visual inspection and materials testing. This is made possible due to the vast amount of experience in constructing buildings using known processes and materials in known environments. Buildings are designed and built to defined standards/codes. Construction on the lunar surface cannot rely on heritage, experience or codes to guide practices. Therefore, it is assumed that a method of verification of as-built structures will need to be employed to verify structure performance prior to putting the structure into service. Concepts must have methods to verify as-built performance to be considered a viable option. Non-Destructive Evaluation and Testing (NDE/NDT) methods and "digital twins" may play a significant verification role during and after construction.

Performance as a Landing/Launch Surface: The primary purpose of a LLP is to reduce risk to the vehicle, crew, and surface/orbital assets. Landing/Launch pads reduce risk by: 1) Providing a known, benign, launch/landing surface. 2) Mitigating plume -regolith interactions that could produce projectiles able to impact the lander and other assets, cratering below the lander, dust plumes that limit the efficacy of sensors and obscure vision and other risks. 3) Providing protection from surrounding assets in case of an anomaly such as a propellant tank explosion or off-nominal launch/landing attempt. 4) Facilitating crew and cargo loading/unloading operations. 5) Providing supporting infrastructure to service vehicles. 6) Providing other supporting infrastructure such as landing aids, beacons, visual landing targets, lighting and other services. A risk assessment can provide quantifiable data for this criterion.

Expected Life: Expected Life is the duration of serviceable life of the LLP. Expected life can be quantified by the number of launch/landings, time until exposure to the lunar environment degrades the performance of the structure to the point of needing a repair, or other factors.

Ease of Repairability: Some concepts may provide repair measures to extend the serviceable life of the LLP. Examples could include re-securing surface stabilization measures, filling "pot holes", replacing damaged pavers, re-compacting regolith and many others. Quantifiable metrics include launch mass for repair materials and systems, power requirements, time for repairs, etc.

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Lifecycle Cost: Lifecycle cost is the total capital investment from project initiation to decommission of the LLP structure. Some concepts may have a higher initial investment and lower subsequent costs. Other concepts will have a lower initial investment and more expenses during the life of the LLP.

LANDING/LAUNCH PAD CONCEPTS

Proposed landing and launch pad concepts span a wide range of methods and components. For the purposes of this paper LLP components include a central impenetrable zone for nominal landings and launches, an outer perimeter for off-nominal landings and dust mitigation, and barriers to mitigate blast ejecta and blast impingement. Subsystems like power, communications, fueling and other Surface Support Equipment (SSE) that are required for a full Launch & Landing Complex are out of scope for this paper. There are many concepts for construction of LLPs with a great deal of variation in maturity. This paper attempts to establish a listing of all LLP construction concepts to date surveyed by the authors. Some concepts are simply ideas, others have undergone some development. Each LLP construction concept is briefly discussed below. The list represents concepts generated by the authors and prior art. Where prior art exists the authors are referenced. We envision that this list will grow over time as new LLP concepts are identified and explored. In a trade matrix, each concept below represents a row against which performance criteria are assessed in the columns of the matrix. In the future, another dimension in the trade matrix may be required to assess each construction concept for use in the LLP central zone, perimeter and barriers.

Table 2. Launch and Landing Pad Concepts

LLP Structure Concepts
Minimal Preparation
Existing Topography
Compacted Regolith Surface
Bedrock Surface
Ice Surface
Rock Piles
Surface Stabilization Applications
Regolith Bags
Ice Bladders
Pavers
Metallic Plates
Deployable Structures
Direct Emplacement of Sintered Structures
Direct Emplacement of Polymer Concretes
Direct Emplacement of a Concrete Pad

Minimal Preparation: Launch and landing occurs directly on the surface with minimal to no site preparation. This solution can be sufficient if the lander/launcher thrust configuration, concept of operations, or overall mission architecture minimizes risks to acceptable levels without the use of a LLP.

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Existing Topography: Site selection, natural surface features, and/or lander configuration may be used to mitigate LL risks.

Compacted Regolith Surface: Surface regolith is either compacted or the loose surface regolith is removed to expose dense, compacted regolith (van Susante, 2016).

Bedrock Surface: Surface regolith overburden is removed to expose bedrock.

Ice Surface: In locations where a consolidated mass of icy material is found exposed at the surface or protected from sublimation by a regolith overburden that can be removed, the ice surface could be used as a LLP after measurements of its characteristics.

Rock Piles: Appropriately sized rocks are collected, transported and emplaced to form LLP structures. The most basic approach is simply to place rocks in the desired locations; however, rock size requirements may make additional stabilization methods a requirement. Stabilization methods may include the use of meshes to constrain rock, gabion cages, strategic placement of rock to employ natural interlocking features, and welding rocks together to create a monolithic structure (van Susante, 2016).

Surface Stabilization Applications: On Earth, surface modification for soil stabilization is used in many applications to minimize dust-related hazards and operational impediments. Examples include mines, construction sites, airfields, helipads, and others. Earth-based solutions may be applicable in some form for lunar surface LLPs. Typical terrestrial methods for soil stabilization include spray application of polymers, inclusion of fibers, emplacement of grids, and others.

Regolith Bags: Bags made from basalt glass fiber or other materials are filled with regolith and emplaced. Regolith bag geometries include traditional sand bags, overlapping "ravioli"-shaped "Moselini" bags, sand tubes, and broad blankets.

Ice Bladders: Bladders are emplaced that are filled with water and allowed to freeze forming an ice structure. Potential bladder shapes include various smaller geometries, overlapping "ravioli"-shaped bags, large cylinders, and broad blankets.

Pavers: Pavers/bricks are produced in-situ and then transported and emplaced into the desired LLP geometry. Pavers are constructed by filling molds with materials and applying one or combined methods to bind the materials together in the mold shape. Binding methods include sintering, addition of polymers, application of pressure, cements and others. Sintered pavers can be constructed using several heating methods including solar, resistive heaters, microwave, laser, induction, and thermite. Polymer bound pavers can employ thermoplastics, two-part and ultraviolet light (UV) cured epoxies, geopolymers, and plant-based biopolymers. A key trade for polymer and cement-based pavers is the origin of the polymers/cement and associated launch mass. Joints between pavers can be managed in several ways including interlocking features, welding via a torch or concentrated solar, polymer, thermite joining. Straps, anchors, and/or wire mesh can be used to constrain paver positions (Kelso, 2016), (Koh, 2010), (Fateri, 2019), (Taylor, 2005), (Gosau, 2012), and (Hobosyan, 2014).

Metallic Plates: Metallic plates are emplaced in the desired LLP geometry and secured together via locking features, welding, fasteners or caulking. Straps, anchors and/or wire mesh can be used to constrain plate positions. The plates (e.g., steel) can be launched from Earth, but this is unlikely until the delivery cost of lunar payloads is lowered to make such a shipment affordable with associated higher launch rates. Alternatively, the plates could ultimately be produced locally from ferroalloys and other alloys produced with processes such as Molten Regolith Electrolysis or Carbothermal Reduction.

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Deployable Structures: Deployable structures include mats/blankets and gossamer structures of mesh or fabric using metallic wire, basalt glass, or other fibers. Some concepts allow for structures to be deployed for launch/landing operations and then retracted during periods of inactivity. Retraction of the structure may provide greater system life by reduction of exposure to the lunar environment. Deployable structures can be constrained to the surface in various ways including drill-type anchors, regolith bags, thermite anchoring processes, polymer anchoring processes, straps, inflatable structural elements, and others.

Direct Emplacement of Sintered Structures: There are several concepts for sintering regolith in place. Most concepts use an automated system to position a heating device above prepared regolith followed by rastering of the thermal energy beam to trace out the desired geometry similar to a Fused Deposition Modeling (FDM) 3D printer. Some concepts include rastering solar concentrators or lasers while others constrain the energy source to a fixed location in a configuration similar to a UV cure or powder bed 3D printing process. A single layer or multiple layers may be emplaced depending on the process and desired geometry. Emplacement process concepts include directly sintering the regolith on the lunar surface, depositing a fine layer of regolith on the surface and then sintering layers repeatedly until a sufficient slab depth is achieved, or depositing a "paste" of regolith subsequently heated to sintering temperatures. Heating methods for sintering in place include concentrated solar thermal energy, resistive heating, induction, microwave, lasers and thermite reactions (Hintze, 2012), (Taylor, 2005), and (Hobosyan, 2014). Selective Separation Sintering (SSS) has also been proposed to create expansion joints by emplacing a sintering inhibitor along a suitably graded area in the shape of a grid. The inhibitor in the interfacial area has a higher sintering temperature than the targeted structural material and therefore aids in counteracting the effect of expansion and contraction of the sintered material due to thermal effects (Khoshnevis, 2015).

Direct Emplacement of Polymer Concrete: A host of concepts using polymer binders with regolith exist. Most of the concepts use an automated system to position an extruder above the location where the slab is to be constructed. The deposition system is rastered to trace out the desired geometry similar to a Fused Deposition Modeling (FDM) 3D printer. A single layer or multiple layers may be emplaced depending on the process and desired geometry. Another approach similar to a powder bed 3D printing process deposits a fine layer of regolith on the constructed surface and then a polymer spray nozzle "wets" the regolith to bind it. The process is repeated until a sufficient slab depth is achieved. An alternative method is to produce a large quantity of regolith/polymer mixture and then pour a slab similar to concrete pouring on Earth. Adequate polymers may include thermoplastics, two-part and UV cured epoxies, geopolymers, and plant-based biopolymers. A key trade for directly emplaced polymer concrete is the origin of the polymers and associated launch mass (Mueller, 2018).

Direct Emplacement of a Concrete Pad: Hydraulic or non-hydraulic concrete formulations tailored for the lunar surface and associated rebar can either be produced from local resources (e.g., sulfur) or imported from Earth and emplaced with modified terrestrial methods. Imported mass may require increased lunar surface payloads and/or higher launch rates and specialized robotic equipment would be required.

TECHNOLOGY GAPS

A detailed discussion of technology gaps will be provided in the future once the trade study has been more thoroughly assessed. However, it is readily apparent upon inspection of the state

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of the art that significant technology gaps exist. Although there has been some work completed on various concepts, all remain in a low TRL status. Emplacement systems, processing/production strategies, materials, and structures must be prototyped and exposed to lunar environmental conditions in order to raise the TRL. Specifically, prototypes must be built and subjected to simulated lunar conditions including thermal vacuums, illumination and radiation including UV, launch/landing environments, reduced gravity, micrometeoroid impacts, regolith surface charges in the environment, and others.

CONCLUSION

NASA is committed to returning humans to the lunar surface by 2024 and creating a long term, sustainable presence for exploration and utilization as early as 2028. Apollo era approaches to landing and launching operations on the lunar surface will result in significant mission risks in this new era. Interactions between landing/launch rocket engine plumes and the unprepared lunar surface will cause regolith to be eroded and ejected at high velocities from beneath the vehicle. The associated ejecta elevates mission risks by: obscuring sensors and human vision during landing via lofted dust; cratering/modifying the surface that the vehicle will land upon; and subjecting the vehicle, surrounding assets, and potentially orbital assets to impacts and contamination from high velocity dust particles. Risks may be mitigated to some degree by new vehicle configurations, specialized landing/launch operations, or mission architectures. However, Launch/Landing Pad infrastructure will be necessary to reduce mission risks to acceptable levels and grow human lunar activities beyond "footprints and flags" into a vibrant scientific and economic endeavor. This is true for the same reasons why airplanes use runways and airports, ships use docks and seaports, cars and trucks use roads and service stations, trains use rails and stations, and Earth-based space operations use launching and landing complexes. Off-Earth spaceports will need Launch and Landing Pads (LLP) and eventually Launch and Landing Complexes to reduce mission risks and facilitate the exploration and utilization of off-Earth resources.

In order to prepare for the need of LLPs this paper establishes a baseline of the current state of the art, provides a listing of known LLP concepts, and presents criteria to trade concepts and identify gaps for future technology development.

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